

The Role of Innovation Ecosystems and Social Capital in Startup Survival

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Abstract—Innovation ecosystems, including technology parks and incubators, endeavor to achieve a high density of startups that is rich in social capital to help tenants evolve. Given the importance of social capital on startup success and the opportunity for social capital diffusion in high-density ecosystems, the inconclusive impact of ecosystems on firm performance warrants further attention. We present a startup survival model that distinguishes between access to social capital and using it. Empirical tests with the longitudinal Kauffman Firm Survey find that those startups that use social capital by collaborating with other agents (universities, industries, and government organizations) significantly outperform startups that do not. However, we also observe that the amount of social capital available in an ecosystem counterintuitively does not correlate with a tenant's use of it, and correlates weakly with longer life expectancy. These results are moderated by whether the tenant is a high-tech startup or not. Findings contribute to extant research in innovation and entrepreneurial ecosystems. Policy and pedagogical recommendations for leveraging an ecosystem to improve firm performance and achieve the ultimate goal of regional economic development are discussed.

Index Terms—Entrepreneurship, innovation ecosystem, social capital, startup collaboration, startup firms, survival analysis.

I. INTRODUCTION

THE role of social capital in entrepreneurship is a prominent research topic whose importance is broadly acknowledged. For example, meta-analyses find social capital impacts new firm performance [1], [2]. One context in which social capital and entrepreneurship are frequently discussed is that of the innovation ecosystem. Innovation ecosystems support startups in their development process in part by helping them build support networks [3]. An innovation ecosystem is defined by the complex relationships formed between actors, whose objective is to enable technology development and innovation thereby growing the economy and creating jobs [4].

One common feature of an innovation ecosystem is the availability of research and commercial resources. A second common feature is that entities in an ecosystem are geographically localized and focused on few and overlapping industries. Examples of

innovation ecosystems are regional clusters, industry parks, and startup incubators, which are characterized by a high density of companies in similar industries collocated in the same ZIP code and managed by universities, municipalities, or states/provinces so as to promote regional economic development [5]–[8]. This paper considers innovation ecosystems operating at this scale.

A high-priority goal of any innovation ecosystem is the development of robust business and social networks [9]. These networks can offer on-site and off-site agents for innovation and commercialization, and include investors, business service providers, university and government labs, and other startups. They can also lead to the sharing of tacit knowledge and building of social capital. A firm's social capital is a measure of its relationships in its ecosystem, relationships that can facilitate coordination and collaboration yielding a competitive advantage [10], [11]. Social capital has been found to correlate with entrepreneurial success when the startup is in an innovation cluster or location [12]–[14] and is more indicative of entrepreneurial success than intellectual property [15].

As important as social capital is, especially in the context of innovation ecosystems, the concept of social capital in entrepreneurship remains fuzzy and our understanding of the processes involved in attaining it is limited [16]. Dominant logic assumes social capital lies in the resources embedded in networks; however, reason would dictate that social capital is not possessed but exists only through participation. Furthermore, because firms collocated in an innovation ecosystem tend to represent a wide range of stages of maturity with different needs, accessible social capital may be different from what is needed leading to a difference in social capital access and social capital utilization. Researchers claim the process of building social capital is too often conflated with outcomes thereby obfuscating its true nature [17] and questions have been raised as to how social capital actually works in practice [18].

Gedajlovic *et al.* [19] criticize researchers for failing to investigate the link between relationships and outcomes. They note the flawed practice of referring to social capital in terms of the nature of the relationships through which resources are *expected* to be derived, not resources actually derived. In sum, although there is extensive literature on entrepreneur social networks, there is relatively limited analysis of the actual social capital that these networks generate [19]. The danger in ignoring possible differences between access to and utilization of social capital is illustrated in Fahey and Prusak's [20, p. 266] critique of the knowledge management discipline. They state the second most common error committed by researchers and

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practitioners is “emphasizing knowledge stock to the detriment of knowledge flow.”

This paper aims to address this gap by investigating the relationship between the availability of social capital in an innovation ecosystem and the performance of firms in the ecosystem, the relationship between the utilization of social capital by these firms and their performance, and distinguishing these two relationships. We contribute to the literature by considering a number of key questions. First, we aim to clarify the concept of social capital in entrepreneurship and our understanding of the processes involved in attaining it, and ask if having access to social capital increases its utilization. Next, we look to explain conflicting findings in the literature on the benefits of ecosystems [21], [22], and ask if having access to and/or utilizing social capital increase the odds of a firm’s survival. Furthermore, we address our understanding of the impact of social capital by industry type, and ask if the processes involved in building and using social capital differ between high-tech and low-tech industries. Finally, we empirically test these relationships using a nationwide longitudinal database, the Kauffman Firm Survey (KFS).

Our findings suggest that benefits offered by innovation ecosystems, in particular collocation and the availability of social capital, do not necessarily lead to startup success. This seems to confirm studies that find that regional clusters or startup incubators may offer conditions conducive to new venture creation but do not necessarily improve their odds of survival. Our study also finds that the availability of social capital does not have a significant relationship with the utilization of social capital. In other words, while ecosystems can offer many benefits, it does not guarantee that firms will actually capitalize on those benefits. Finally, we find a strong positive relationship between the utilization of social capital and survivability, more so for high-tech firms than for low-tech firms.

The remainder of the paper is organized as follows. The next section will review the relevant literature and develop our model and hypotheses. Subsequently, methodology including data collection, construct measurement, and analysis steps are discussed. Empirical results are then presented followed by a discussion of findings and conclusions.

II. MODEL DEVELOPMENT AND HYPOTHESES

Innovation ecosystems are an increasingly popular concept that describes the networked and systemic nature of innovation. The concept focuses not only on innovation activities but also on the interaction and interdependency among actors [23]. It integrates invention with commercialization, and includes profit-driven systems and policy that supports innovation and economic development. As such, the term innovation ecosystem has begun to describe more traditional spaces such as clusters, incubators, and technology parks [23].

Geographic concentrations of interrelated companies in a particular field compete but also cooperate [5]. In the organization-of-markets literature, it has become an accepted wisdom that similar businesses will cluster in physical space [5], [24], [25] because firms can benefit from an ecosystem and access to resources, specialized staff, venture capital, suppliers, and support

services [26]. Proximity also has the benefit of stimulating communication and the scientific exchange of ideas [27].

Research found that geographic clustering or startup density can lead to performance benefits. For example, Audretsch and Feldman [28, p. 31] argue that one of the greatest insights in innovation is that geography matters, and that “a long tradition of analyzing the innovative process within the boundaries of the firm and devoid of spatial context has given way to the incorporation of spatial context.” Mainstream literature considers geographic proximity to be a driver of innovation [29]. Theories as to why geography matters include that the concentration of firms improves the production efficiency and allows easy access to needed resources, a highly competitive environment can encourage firms to be proactive and quickly build needed competencies [30], and knowledge flows are particularly strong within the same region [31].

Technical knowledge spillovers from collocating were found to help entrepreneurs learn about potential partners and opportunities [32] and positively influence product innovation [24]. In their work on the knowledge spillover theory of entrepreneurship, Audretsch and Keilbach [33] find that regions with a high investment in knowledge tend to experience high knowledge spillover, while regions with low knowledge investment experience low spillover. They also found that a high knowledge context generates new ideas whereby entrepreneurial opportunities are then created when spillover is exploited.

Innovation ecosystems are also supported by a positive “clustering effect” because firms with shared ambitions are in close proximity to each other [34]. Clustering might facilitate effective networking and knowledge spillovers between firms with similar interest, knowledge, and expertise [34]. In a review of incubation literature, McAdam and Marlow [34] found key themes were the provision of physical infrastructure, access to professional networks, and a clustering effect. A regional cluster is similarly defined as a geographic concentration of interrelated actors in a particular industry that supports entrepreneurship and innovation [30]. More precisely, it is a “critical mass of local knowledge, expertise, personnel, and resources used by firms to gain competitive advantages” [35, p. 428].

The literature, however, is not clear whether the benefits of localized ecosystems are in fact realized. For example, researchers and practitioners have published case studies and best practices for the management of innovation ecosystems, but verifying their economic impact remains elusive [6], [36], [37]. Incubators rarely track their tenants once they graduate from the incubator and into a less supportive environment [38], [39], and a cohort study by Schwartz [21] reveals that startup survival is statistically significantly better among companies that never participated in an incubator. Among clusters, Ruland [22] finds that the profitability of smaller firms tends to be considerably lower than for firms that opt not to join clusters.

The above research investigates the association between the networking opportunities in an innovation ecosystem, including opportunities for innovation and collaboration, and the performance of new ventures in the ecosystem. Another line of research into the association between networking and new venture performance is that of social capital. Literature defines social capital as a company’s relationship with its entire ecosystem

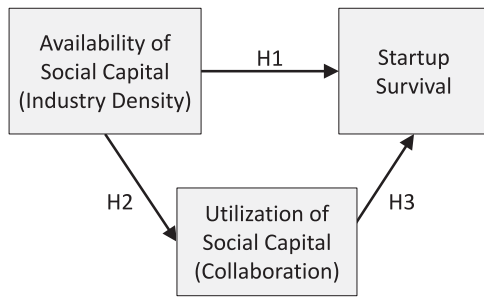


Fig. 1. Conceptual framework of the study.

and one type of intangible asset that forms its intellectual capital [10], [11], [40]. Social capital's central tenet is that networks of relationships constitute a valuable resource. Social capital correlates with entrepreneurial intention [41] and can correlate with entrepreneurial success when the startup is in an innovation ecosystem [13], [42]–[44]. Moreover, relationships have been found to be more indicative of entrepreneurial success than intellectual property [45].

The literature on innovation ecosystems discusses the availability of resources with which new ventures can build social capital, whereas social capital literature discusses the use of these assets by new ventures. If we assume that the availability of social capital leads to the benefits of social capital, we can propose the following.

Hypothesis 1: Social capital availability is positively associated with startup survival.

However, the availability of an asset and the use of that asset are two separate things, and perhaps it is the assumption of the latter by innovation ecosystem research that has led to its inconclusive findings. Battisti and McAdam's study of graduate entrepreneurs [46] found that while networks of external professionals and advisers were accessible and valued, they were not relied upon. Therefore, it appears that access to social capital may not necessarily lead to its utilization. This may be because although social capital is available, it may be difficult for entrepreneurs to locate the right individual in a complex network [3]. In addition, Shaw [47] found that although social capital can provide new firms with access to a diverse set of resources, firms need to be motivated to access these resources and patterns of utilization can be complex. Moreover, Tornikoski and Newbert [48] found that actively networking and receiving outside help led to emerging entrepreneurial firms. Social capital does not only need opportunity, it also needs motivation and ability [49]. We therefore propose that the utilization of social capital is a mediating factor between access to social capital and startup survival, which yields the following additional hypotheses and the overall model in Fig. 1.

Hypothesis 2: Social capital availability is positively associated with social capital utilization.

Hypothesis 3: Social capital utilization is positively associated with startup survival.

Innovation ecosystems matter when firms need access to natural resources, other firm's complementary skills, and a competent labor supply [29]. Benefits are clear in low-tech industries, where supply chain costs are substantial (e.g., transporting

supplies and finished goods) and firms in an ecosystem can benefit from (shared) economies of scale. But what about high-tech industries, where the dissemination and access to knowledge and resources are seemingly less constrained by geography? In this case, we would not expect geographic location to limit the extent of social capital and exchange of tacit knowledge. However, in their study on the biotech industry, Stuart and Sorenson [25] found clustering plays an important role in high-tech industries because the social relationships give startups access to needed resources.

Empirical studies of high-tech firms found that information exchange does not always occur formally thereby highlighting the importance of informal communication and collocation [16]. Although high audiovisual quality teleconferencing is becoming ubiquitous, informal communication can still be difficult; emotion and body language are not well conveyed, participants must coordinate actions more precisely, and by not facilitating low-latency interaction, it can impede important spontaneous discussion [50].

Research suggests that social capital offers a complex system of relationships that are relevant to both low-tech and high-tech industries and social capital can be quickly accumulated in both [51]. In a pivotal study, Liao and Welsch suggest that high-tech entrepreneurs seek novel information and tacit knowledge, whereas low-tech entrepreneurs seek physical resources [52]. The information acquired through social capital may benefit high-tech startups more than low-tech firms because social capital has been shown to correlate more strongly with the performance of firms that operate in "uncertain environments where knowledge is widely dispersed and rapidly evolving" [2, p. 17], which is more characteristic of high-tech firms than of low-tech firms. Social capital also helps technology entrepreneurs conduct opportunity recognition and cross the "valley of death," two activities critical to high-tech firm survival [53]. We would thus expect

Hypothesis 4: Social capital utilization is more positively associated with survival among high-tech startups than among low-tech startups.

III. METHODOLOGY

A. Data

This study uses the confidential version of the longitudinal KFS of 4928 companies founded in 2004 and surveyed annually from 2004 to 2011 [54]. The KFS collects information about each participating firm's business characteristics, strategy and innovation, business organization and human resource benefits, business finances, work behavior, and ownership and demographics of its active owner and operators (not silent investors).

To facilitate longitudinal survival analysis and reduce the impact of missing data, this study uses the weighted imputed long format KFS dataset "Longitudinal_Long_MI_Survival_Ready" hosted by the National Opinion Research Center Data Enclave for the Kauffman Foundation. A detailed description of the structure and contents of this dataset is provided by Farhat and Robb [55]. The weights remove response bias from attrition and selection bias among high tech, medium tech, low tech, and woman-owned firms. Imputation creates five versions

TABLE I
AVERAGE INDUSTRY DENSITY BY STARTUPS' 2004 NAICS SECTOR CODE

# of Startups*	Average Density	NAICS Sector ID	NAICS Sector Title
1203	128.8	54	Professional, Scientific, and Technical Services
243	114.4	45	Retail Trade (General Merchandise, Non-Store Retail)
283	114.2	44	Retail Trade (Market Specific Retail)
121	111.3	62	Health Care and Social Assistance
186	73.7	52	Finance and Insurance
98	68.2	72	Accommodation and Food Services
390	67.4	23	Construction
454	64.6	81	Other Services (except Public Administration)
219	58.9	42	Wholesale Trade
518	42.0	33	Manufacturing (Mechanical, Electrical)
151	39.2	32	Manufacturing (Lumber, Chemicals, Pharma)
177	38.1	53	Real Estate Rental and Leasing
360	37.8	56	Admin, Support, Waste Mgmt & Remediation Services
51	35.0	31	Manufacturing (Food, Beverage, Textile, Apparel)
102	23.9	48	Transportation and Warehousing
164	20.3	51	Information
10	20.2	49	Transportation and Warehousing (Messenger & Storage)
105	12.2	71	Arts, Entertainment, and Recreation
29	9.9	61	Educational Services
10	4.4	55	Management of Companies and Enterprises
40	2.3	11	Agriculture, Forestry, Fishing and Hunting

*Statistics on fewer than ten KFS firms are not publicly disclosed in accordance with KFS confidentiality policy.

of the original dataset with values for missing data that are generated following their statistical distribution for that firm (multiple imputation), and last-observation-carried-forward or last-observation-carried-backward corrections to questions that the participant declined to answer (logic imputation of soft missing data). The variables used in this study are not impacted by multiple imputation; logic imputation reduces the percentage of soft missing values to under 2% for collaboration and under 0.4% for North American Industry Classification System (NAICS) (used to calculate industry density).

We augmented the KFS dataset with the County Business Patterns (CBP) dataset of the United States Census Bureau. For every firm and year in the KFS up to 2010 inclusively, the augmented dataset provided the number of companies in its ZIP code with the same NAICS sector code. Finally, the KFS asked participants if they considered their firm to be high tech or not. This study classifies firms that responded affirmatively as high tech.

B. Measures

1) *Access to Social Capital*: Social capital is often categorized as strong intragroup ties and weak intergroup ties, with researchers treating the dynamics of each category distinctly [56], [57]. However, the frenetic pace of collaborations, pivots, and reorganizations that startups can experience, for example, in which competitors become partners and team members become spin-offs, calls into question this categorization [58], [59]. For this study, we use a startup's industry density of potential relationships as a proxy for access to social capital. It is calculated

using the number of companies in the startup's five-digit ZIP code with the same NAICS sector code. Filtering the number of firms in the startup's ZIP code by the startup's NAICS code yields a density measure that is specific to the startup's industry. Industry density includes potential competitors, but a ZIP code with a higher density measure could be expected to include a greater number of potential collaborators as well.

The five-digit ZIP code was used because, of the geographical standards for which industry density can be computed using census data, it is the closest in size to that of an incubator or cluster. Alternative geographical standards, namely county, metropolitan statistical area (MSA), and state, are much larger than incubators and clusters. An analysis of the United States Postal Service ZIP code list (current as of March 2017) shows that, on average, there are 15.5 ZIP codes per county, 100 ZIP codes per MSA, and 576 ZIP codes per state.

Overall, KFS firms subscribed to 24 different NAICS sectors when they formed in 2004 (see Table I). Industry density varies significantly among sectors; for example, industries in agriculture, forestry, fishing, and hunting (NAICS sector 11), which are traditionally rural, have the lowest density. The zip codes of 98 KFS startups did not match CBP records; these firms were excluded from density analyses. Over the duration of the KFS, 513 distinct startups reported that they were a high-tech firm.

As to be expected, industry density also varies significantly across geographical regions (see Table II). Every year at least 75 KFS startups were the only firms of their NAICS sector in their ZIP code (density = 1), whereas several other KFS startups were one of over 800 establishments in their same ZIP code and NAICS sector (density >800). Examining the geographical

TABLE II
DESCRIPTIVE STATISTICS OF DENSITY PER YEAR AMONG ALL
SURVIVING STARTUPS

Year	Density Mean	Density Std. Dev.	Density Min	Density Max
2004	73.7	94.2	1	991
2005	72.0	92.8	1	953
2006	75.2	93.3	1	825
2007	77.5	97.3	1	934
2008	77.2	100.0	1	902
2009	74.1	96.5	1	896
2010	73.6	94.3	1	866

TABLE III
NUMBER OF KSF STARTUPS INHABITING THE SAME ZIP CODE

Percentile	10%	25%	50%	75%	90%	95%	100%
KFS Startups/ZIP	≤1	≤2	≤5	≤8	≤9	≤12	≤41

clustering of KFS firms offers an observation of spatial variability. During the years in which they were tracked, KFS firms inhabited 3438 distinct 5-digit ZIP codes, over half of which hosted 5 KFS startups or more, 288 ZIP codes hosted 10 startups or more, and 4 ZIP codes hosted 25 startups or more (see Table III).

The industry density of a KFS startup also varies temporally between 2004 (when they formed) and 2010 (the last year of CBP data) with a standard deviation of on average 19.47, which is significant when compared to the mean density. Variation in a startup's industry density is due to several reasons: 24.8% of the firms pivoted to a different NAICS sector, 30.9% moved to a different ZIP code, other companies with the same NAICS sector may have moved into or out of the startup's ZIP code, and companies in the startup's ZIP code may have pivoted into or out of the startup's NAICS sector. Fig. 2 plots 2004 versus 2010 density of all the KFS firms that survived to 2010. While most densities did not change, there is significant variability ($R^2 = 0.5549$) to indicate the use a longitudinal startup model with density as a time-varying covariate.

Industry density is different from resident population, even though both are attributes of the ZIP code in which a startup resides. The former is a count of firms with the same NAICS sector as the startup, whereas the latter is a count of residents independent of any industry. For the startup, the former represents a potential source of relevant social capital, whereas the latter does not enter into our model. To visualize the difference between industry density and resident population, we generate a two-dimensional histogram of the industry densities and populations of the ZIP codes occupied by all KFS companies during the years they are tracked (see Fig. 3). The size of each histogram bin is 50 companies in the same ZIP code with the same NAICS sector (vertically) by 5000 residents in the same ZIP code (horizontally), and the area of a bubble is proportional to the total number of times from 2004 to 2010 that a company exhibits the bubble's coordinates. A visual inspection of this histogram shows that industry density and population are not strongly correlated; they share just $R^2 = 0.082$ of their variance.

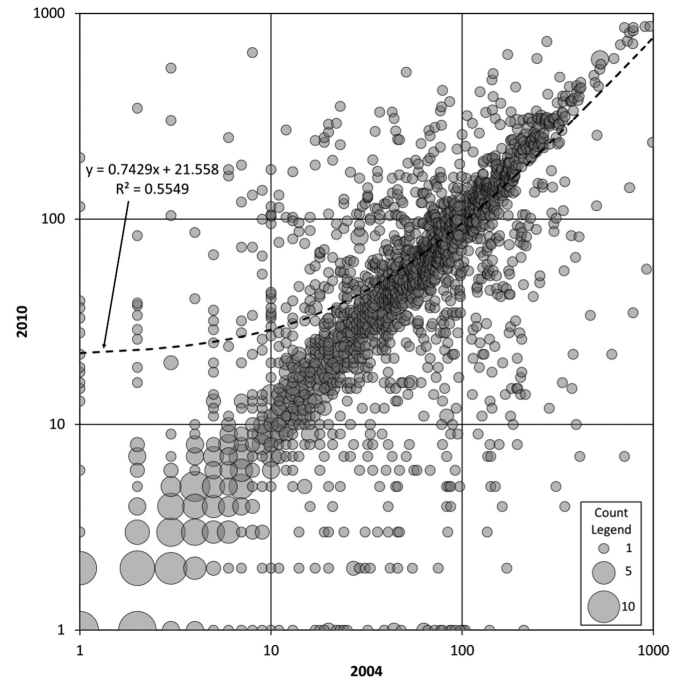


Fig. 2. Changes in industry density of KFS startups between formation in 2004 and 2010.

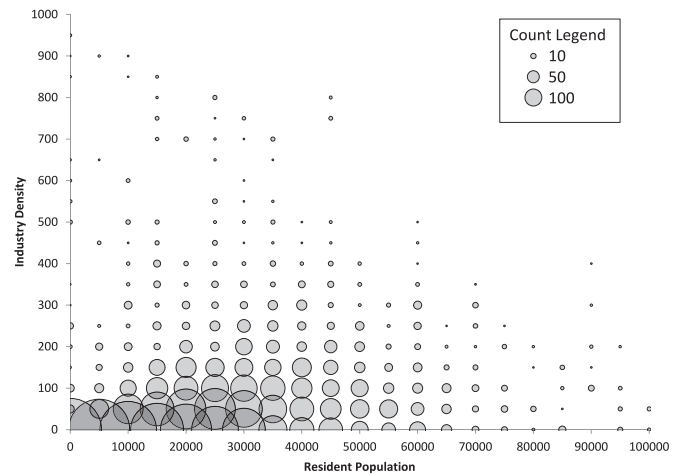


Fig. 3. Histogram of startup locations by industry density and population.

2) *Utilization of Social Capital:* Literature claims that successful entrepreneurs build social capital and tacit knowledge through collaboration. Therefore, a startup's use of social capital can be measured by counting the collaborations that contribute to the startup's success. These collaborations are with agents offering complementary resources requisite for commercialization, including intellectual property, production, sourcing, distribution, marketing, and/or financing. This measure of a startup's use of social capital is distinct from the startup having access to social capital (i.e., industry density). Note that to test the hypotheses one need not consider collaborations exclusively within a specific geographical region, as with industry density (i.e., the ZIP code). By not constraining collaboration geographically, we evaluate further the significance of density.

TABLE IV
COUNT OF COLLABORATING STARTUPS AND PERCENT OF COLLABORATING
SURVIVORS BY TYPE OF PARTNER

Year	Companies		Universities		Gov. Labs	
2007	328	14.08%	81	3.48%	40	1.72%
2008	318	15.20%	69	3.30%	34	1.63%
2009	261	13.54%	76	3.94%	29	1.50%
2010	223	12.56%	62	3.49%	29	1.63%
2011	198	12.04%	52	3.16%	24	1.46%

Starting in the 2007 survey, the KFS asks founders whether or not they believe that collaborations with other companies, universities, or government labs improved their firm's competitive advantage [55]. This data does not describe the nature of the firm's network in detail, such as the topology of strong and weak ties [60], [61], other than classifying collaborative partners into the three categories (other companies, universities, or government labs). However, network information may not be necessary if the self-reported claim that the collaboration impacted the firm's competitive advantage is trustworthy.

When measured as a percentage of surviving companies, collaborations are relatively consistent (see Table IV). This cross-sectional analysis, however, does not distinguish between few companies collaborating consistently and many startups collaborating at different years. If the latter, a startup survival model would use collaboration as time-varying covariates; if the former, the model would use collaboration as a fixed covariate. Thus, the startups' patterns of collaboration activity must be inspected before selecting a model.

Collaboration activity can be represented as a five-bit dummy variable, with the least significant digit being "1" if the company collaborated in 2007, the digit to the left being "1" if the company collaborated in 2008, and so forth with the most significant digit being "1" if the company collaborated in 2011. For example, the pattern 00101 indicates the company collaborated in 2007 and 2009 and did not collaborate in 2008, 2010, nor 2011.

Table V presents the patterns of collaboration with the three types of collaborative partners (the pattern 00000 is excluded). The column "Count" indicates the number of KFS startups that exhibited that pattern of collaboration, and patterns are displayed in the order of decreasing count. Of the 3140 KFS startups operating in 2007, 706 collaborated at least one year with another business, 196 collaborated at least one year with a university, and 99 collaborated at least one year with a government laboratory. These numbers are more than double to those of Table IV, indicating that a startup's collaboration activity varies from year to year; this conclusion is also apparent from the five-digit patterns themselves.

We thus treat collaboration as a time-varying covariate in our model, and represent it for each year t of each startup i as the sum of its types of collaborations $x_{C,i,t} \in [0, 1, 2, 3]$. For example, if the founder of the 400th firm in the KFS reports in 2010 that collaboration with companies and universities – but not government labs – yielded a competitive advantage, then $x_{C,400,2010} = 2$.

TABLE V
PATTERNS OF COLLABORATION

Patterns of Collab with Company	Count	Patterns of Collab with University	Count	Patterns of Collab with Government	Count
000001	117	000001	36	000001	22
000010	90	000100	23	000010	15
000100	60	000010	20	001000	12
000011	51	001000	18	010000	12
001000	48	010000	16	000100	7
010000	39	000011	13	000011	4
000111	22	011111	10	000111	3
000110	19	001100	7	011100	3
011000	19	000110	6	000101	2
001111	18	000111	6	001011	2
001110	17	011100	6	001100	2
000101	15	000101	5	001110	2
011111	15	011000	5	001111	2
001001	13	010111	4	010101	2
001011	13	001001	3	011110	2
010100	13	011101	3	001001	1
011110	13	001010	2	001010	1
001100	12	001111	2	010100	1
010001	12	010001	2	010110	1
011100	12	001011	1	010111	1
⋮	⋮	⋮	⋮	⋮	⋮
011001	3	011110	1	011111	1
Total:	706	Total:	196	Total:	99

Consecutive digits from right to left represent collaboration in years 2007 through 2011.

3) *Startup Survival*: Empirical studies into the relationship between social capital and entrepreneurship have used financial and nonfinancial performance measures as dependent variables, including revenue, profit, valuation, employment, and intent (the latter commonly used by studies in entrepreneurship education) [2], [62]. These measures, and their profiles over time, vary significantly between industries and business models, complicating the generalization of such studies. This study uses as the dependent variable a more fundamental and generalizable performance measure: survival. Survival is estimated from the longitudinal KFS using industry density (access to social capital) and collaboration (utilization of social capital) as time-varying covariates.

C. Analysis

We analyze the interactions of industry density and collaboration with startup survival using the Cox nonparametric proportional hazards model for several reasons. First, this survival analysis method does not force us to assume that survival over time follows any particular profile, as would a parametric survival model. Second, we are interested in the effects of the covariates on survival, and thus in a model that generates a hazard ratio for each covariate. The hazard ratio of a covariate has the following useful interpretation: Z one unit increase of the covariate decreases the probability of firm failure by $(1 - \text{hazard ratio}) \times 100\%$. Lastly, the Cox model allows us to employ standard regression techniques.

The Cox proportional hazard regression model expresses the hazard function of startup i at year t as follows:

$$h_i(t) = h_0(t) \times \exp(\beta_D x_{D,i,t} + \beta_C x_{C,i,t} + \beta_H x_{H,i,t}) \quad (1)$$

TABLE VI
EXPLORATORY COX REGRESSION

Covariate	Hazard Ratio	
Collaboration	$HR_C = .690971^{**}$	LR $\chi^2(7) = 147.72$ Prob > $\chi^2 = 0.0000$ log likelihood = -80455
Density	$HR_D = 1.000475^{**}$	
Collaboration $\times$ Density	.999854	
Hightech	.979589	
Collaboration $\times$ Hightech	.735087*	
Density $\times$ Hightech	.998611**	
Collaboration $\times$ Density $\times$ Hightech	1.002028*	

Model passes the LINKTEST fit test; the linear predicted value is statistically significant ($p < .001$) and predicted value squared is not [64]. * $p < .05$, ** $p < .001$.

where $x_{D,i,t}$ and $x_{C,i,t}$ are the industry density and types of collaboration of firm i at year t , $x_{H,i,t}$ is the Boolean variable indicating if firm i reported in year t was a high-tech firm, β_D , β_C , and β_H are the corresponding covariate coefficients, and $h_0(t)$ is the baseline hazard function defined as the hazard function when all covariates are set to zero. The hazard ratios for density, collaboration, and firm type are then $HR_D = e^{\beta_D}$, $HR_C = e^{\beta_C}$, and $HR_H = e^{\beta_H}$, respectively.

Firm failures coincide in the KFS dataset because the KFS is an annual survey and it records events with year-resolution. The Efron approximation for tied failures is used to account for this in the Cox regression [63], which is conducted in STATA v14.

IV. RESULTS

An exploratory Cox regression with all KFS variables discussed and their products provides initial insights into their interactions and effects on startup survival (see Table VI). Collaboration has a meaningful effect on survival; HR_C is not close to 1, so a unit change in collaboration correlates with a significant change in the probability of firm failure. This effect is statistically significant. Moreover, whether a startup classifies itself as a high-tech firm or not (“Hightech” in Table VI) moderates the effect of collaboration on survival in a meaningful and statistically significant way.

Density correlates statistically significantly with the firm survival, but this effect is weak ($HR_D \approx 1$). Thus, a meaningful change in survival correlates only with a large change in density. High tech also moderates this effect. Interestingly, $HR_D > 1$ indicates that a higher density correlates with the increased failure among low-tech firms (Hightech = 0), whereas the hazard ratio of less than one for Density $\times$ Hightech indicates this effect may be reversed among high-tech firms.

The Cox regressions for just low-tech firms and for just high-tech firms are presented in Table VII. A Vuong test confirms that both these models, which use only industry density and collaboration as covariates, fit their respective populations better than the exploratory regression [65]. Collaboration correlates with a reduced risk of failure among both high-tech and low-tech startups, with high-tech firms benefiting more than low-tech firms by a further 10.2% reduction in risk (.685752–.583278).

Density correlates with an increased risk of failure among low-tech startups, and a reduced risk of failure among high-tech startups. This effect is weak. For example, the density x required for a 25% reduction in hazard among high-tech startups implies $e^{\beta_D x} = .75 = (HR_D)^x$, $x = \ln(.75)/\ln(.999266) = 392$

TABLE VII
COX REGRESSIONS BY TYPE OF STARTUP

Low Tech

Covariate	Hazard Ratio
Collaboration	$HR_C = .685752^{**}$
Density	$HR_D = 1.000465^{**}$

LR $\chi^2(7) = 84.34$

Prob > $\chi^2 = 0.0000$

log likelihood = -70274

High Tech

Covariate	Hazard Ratio
Collaboration	$HR_C = .583278^{**}$
Density	$HR_D = .999266^{**}$

LR $\chi^2(7) = 37.13$

Prob > $\chi^2 = 0.0000$

log likelihood = -7187

Model passes the LINKTEST fit test; the linear predicted value is statistically significant ($p < .001$) and predicted value squared is not [64]. ** $p < .001$.

other firms operating in the same ZIP code with the same NAICS sector. This exceeds the density of 97.9% of all high-tech startups in the KFS across all years. Similarly, the density required for a 25% increase in hazard among low-tech startups is $\ln(1.25)/\ln(1.000465) = 480$, which exceeds the density of 99.2% of all low-tech KFS startups. Two simple regressions between density and collaboration, one considering just high-tech startups and the other considering just low-tech startups, reveal that density and collaboration are uncorrelated (adjusted $R^2 < .001$ in both cases).

The exploratory and the two type-specific (high/low-tech) Cox hazards models all support the third and fourth hypotheses; collaboration (a measure of social capital utilization) is positively associated with survival, more so among high-tech startups. The models also support the first hypotheses but only for high-tech startups; industry density (a measure of social capital availability) is positively associated with survival among high-tech startups, but negatively associated with survival among low-tech startups. This density-survival association is weak, and is meaningful only in large industry densities. No support was found for collaboration mediating the effect between density and survival. Consequently, there is no support for the second hypothesis.

V. DISCUSSION

Innovation ecosystems attract new ventures with resources such as lower cost facilities, business services, and opportunities for collaboration. However, we found the availability of social capital does not correlate with startup survival. We also found that the availability of social capital in an ecosystem does not correlate significantly with the utilization of social capital by a startup in that ecosystem. Dense ecosystems can offer the benefits of clustering, but that does not guarantee that firms will capitalize on those benefits. In other words, while dense ecosystems may offer opportunities for collaboration and social capital building, entrepreneurs who chose to conduct these activities will do so regardless of where they are, whereas those who do not will endure on average worse performance also regardless of where they are. Finally, we find a strong positive relationship between the utilization of social capital and startup survival, more so among high-tech firms than among low-tech firms.

There are a number of reasons why social capital may be available and not utilized. Letaifa and Rabeau [29] found that geographic proximity did not always facilitate collaboration, but that social proximity was crucial to both collaboration and innovation. Social proximity is rooted in social embeddedness literature and is defined as relational trust based on friendship and mutual commitment. A lack of social proximity therefore may prevent utilization. This seems to be supported by other research that finds entrepreneurs may not have the specialized skills needed to capitalize on their network ties [46] or the necessary social competence, i.e., the ability to interact effectively with others [66]. Entrepreneurs may also have cognitive biases, such as overconfidence or an illusion of control, which can convince them that they do not need to reach out [40], [67], [68].

Another reason could be that startups need both the opportunity to access resources and the ability to leverage them through a complementary capability, namely absorptive capacity [69]. One of the benefits of social capital is the potential knowledge that can be gained by entrepreneurs. However, experience in collaboration was found to increase the acquisition of new network knowledge [70] meaning education may be necessary. It could also be that the geographical proximity itself is keeping entrepreneurs from utilizing social capital. Even when actors acknowledge collaboration is a win-win situation, the fear of competition may outweigh the benefits of cooperation especially in high-rivalry contexts [29].

Finally, Stuart and Sorenson [25] suggested that although entrepreneurs may prefer to establish new firms in innovation ecosystems, the most productive new ventures are not located in regional clusters. They speculate this may be due to the highly competitive environment that exists in geographically concentrated locations. They also speculate that the benefits from clustering may disappear as the geographic reach of a firm's social network expands.

There are also several reasons why utilizing available social capital correlates differently with the performance of high-tech and low-tech firms. Liao and Welsch find that "sparse, disconnected social networks coupled with strong social ties lead to greater growth aspiration for technology-related entrepreneurs. By contrast, extensive network ties with strong social ties are positively related to the growth aspiration of nontechnology-based entrepreneurs" [52, p. 166]. Thus, the strategic differences between growing a high-tech and low-tech venture, such as prioritizing intangible assets versus tangible assets respectively, impact how and where social capital is sought. Moreover, because the uncertainty of high-tech entrepreneurship, including unproven innovation concepts and business models, is often greater than that of low-tech entrepreneurship, so is the significance of social capital to the high-tech entrepreneur seeking others' tacit knowledge to help manage this uncertainty [2].

This paper has important practical implications for entrepreneurs, innovation ecosystem administrators, policy makers, and educators. Entrepreneurs starting a new venture need to take the time to develop the skills necessary to exploit social capital. Contributing to best practices in managing innovation ecosystems, this paper highlights the importance of social capital utilization to long-term startup success and suggests that

administrators promote collaboration, particularly among high-tech firms. This is supported by Singh [31] who argues for "the deliberate cultivation of interpersonal networks" as a tool for encouraging knowledge flow and innovation. This paper also addresses the concerns of some researchers that the economic impact of managed innovation ecosystems is limited to the short-term benefits of its tenants [21], [22].

Managed ecosystems often promote events to build human capital, such as workshops on accounting or intellectual property protection [7]. Our study does not discourage this practice, but encourages incubator administrators to also leverage and promote the social capital stock they have already built among their tenants. Unlike workshops, which are a one-on-many activity (one subject matter expert to many tenants), building social capital is a many-on-many and more scalable activity.

Although self-organized clusters of entrepreneurs like those found in Silicon Valley are often cited as examples of success, trying to duplicate these through government policy is often elusive. In self-organized clusters, firms connect voluntarily driven by knowledge spillover and often require social proximity from the start. In contrast, clusters created by policy often rely solely on geographic proximity and lack social proximity [29]. Therefore, a focus on building and promoting social proximity as a matter of policy is worthwhile.

A limitation of this paper is that the selection of proxies for the availability and utilization of social capital may affect the internal validity. When responding to KFS questions on collaboration, participants may not have considered less formal interactions with individuals outside the firm, which may have built social capital and even provided a competitive advantage but not yielded formal business collaborations. Similarly, tacit knowledge that yields a competitive advantage can be acquired from distant agents. These two dynamics are not reflected in the density or KFS collaboration measures used in this paper.

VI. CONCLUSION

This paper focuses on the distinction between social capital availability and social capital utilization, and their relationship with firm success. By doing so, our model laid the groundwork for further theory development and empirical research. Although literature tends to assume that geographic proximity is a driver of innovation [29], understanding why some firms utilize the social capital available to them while others do not is an important question. The answer will undoubtedly lie in part within the personality traits of the individual entrepreneur. For example, researchers have used the five factor model of personality traits (openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism) to model entrepreneurs and the performance of their firms [71]–[73]. Several of these personality traits influence not only an entrepreneur's willingness to interact with others but also the willingness of others to interact with her/him.

This paper also has pedagogical implications. Group projects notwithstanding, entrepreneurship education rarely emphasizes the importance of tacit knowledge and the building of social capital for opportunity discovery and risk mitigation.

Instructors need to teach students and practitioners how to develop social capital. By definition, such activities must transcend the classroom and force the learner to engage in a broader ecosystem, just as incubator/cluster tenants must transcend workshops and engage the social capital of their environment.

REFERENCES

- [1] H. Westlund and F. Adam, "Social capital and economic performance: A meta-analysis of 65 studies," *Eur. Plan. Stud.*, vol. 18, no. 6, pp. 893–919, 2010.
- [2] W. Stam, S. Arzlanian, and T. Elfring, "Social capital of entrepreneurs and small firm performance: A meta-analysis of contextual and methodological moderators," *J. Bus. Venturing*, vol. 29, no. 1, pp. 152–173, 2014.
- [3] H. Tötterman and J. Sten, "Start-ups: Business incubation and social capital," *Int. Small Bus. J.*, vol. 23, no. 5, pp. 487–511, 2005.
- [4] D. J. Jackson, "What is an innovation ecosystem," *Nat. Sci. Found., Alexandria, VA, USA*, vol. 1, 2011.
- [5] M. E. Porter, "Location, competition, and economic development: Local clusters in a global economy," *Econ. Develop. Quart.*, vol. 14, no. 1, pp. 15–34, 2000.
- [6] M. Van Geenhuizen and D. P. Soetanto, "Science parks: What they are and how they need to be evaluated," *Int. J. Foresight Innov. Policy*, vol. 4, nos. 1/2, pp. 90–111, 2008.
- [7] C. Colbert, D. Adkins, C. Wolfe, and K. LaPan, *Best Practices in Action - Guidelines for Implementing First-Class Business Incubation Programs*, Revised 2nd ed. Orlando, FL, USA: Nat. Bus. Incubator Assoc., 2010.
- [8] H. M. Al-Mubarak and M. Busler, "Business incubation as an economic development strategy: A literature review," *Int. J. Manage.*, vol. 30, no. 1, pt. 2, pp. 362–372, 2013.
- [9] C. E. Cooper, S. A. Hamel, and S. L. Connaughton, "Motivations and obstacles to networking in a university business incubator," *J. Technol. Transfer*, vol. 37, no. 4, pp. 433–453, 2012.
- [10] R. J. Baker, *Mind Over Matter: Why Intellectual Capital is the Chief Source of Wealth*, Hoboken, NJ, USA: Wiley, 2008.
- [11] M. Feldman and T. D. Zoller, "Dealmakers in place: Social capital connections in regional entrepreneurial economies," *Reg. Stud.*, vol. 46, no. 1, pp. 23–37, 2012.
- [12] S. Y. Cooper and J. S. Park, "The impact of incubator organizations on opportunity recognition and technology innovation in new, entrepreneurial high-technology ventures," *Int. Small Bus. J.*, vol. 26, no. 1, pp. 27–56, 2008.
- [13] P. Davidsson and B. Honig, "The role of social and human capital among nascent entrepreneurs," *J. Bus. Venturing*, vol. 18, no. 3, pp. 301–331, 2003.
- [14] R. Ferguson and C. Olofsson, "Science parks and the development of NTBFs—location, survival and growth," *J. Technol. Transfer*, vol. 29, no. 1, pp. 5–17, 2004.
- [15] J. C. Hayton, "Competing in the new economy: the effect of intellectual capital on corporate entrepreneurship in high-technology new ventures," *R&D Manage.*, vol. 35, no. 2, pp. 137–155, 2005.
- [16] A. Anderson, J. Park, and S. Jack, "Entrepreneurial social capital: Conceptualizing social capital in new high-tech firms," *Int. Small Bus. J.*, vol. 25, no. 3, pp. 245–272, 2007.
- [17] E. A. A. McKeever and S. Jack, "Entrepreneurship and mutuality: Social capital in processes and practices," *Entrepreneurship Reg. Develop.*, vol. 26, nos. 5/6, pp. 453–477, 2014.
- [18] I. D. Light and L. P. Dana, "Boundaries of social capital in entrepreneurship," *Entrepreneurship Theory Pract.*, vol. 37, no. 3, pp. 603–624, 2013.
- [19] E. Gedajlovic *et al.*, "Social capital and entrepreneurship: A schema and research agenda," *Entrepreneurship Theory Pract.*, 2013, vol. 37, no. 3, pp. 455–478.
- [20] L. Fahey and L. Prusak, "The eleven deadliest sins of knowledge management," *California Manage. Rev.*, vol. 40, no. 3, pp. 265–276, 1998.
- [21] M. Schwartz, "A control group study of incubators' impact to promote firm survival," *J. Technol. Transfer*, vol. 38, no. 3, pp. 302–331, 2013.
- [22] W. Ruland, "Does cluster membership enhance financial performance?" *I-Business*, vol. 5, no. 1, pp. 1–11, 2013.
- [23] P. Ritala and A. Almpapoulou, "In defense of 'eco' in innovation ecosystem," *Technovation*, vol. 60, pp. 39–42, 2017.
- [24] B. A. Gilbert, P. P. McDougall, and D. B. Audretsch, "Clusters, knowledge spillovers and new venture performance: An empirical examination," *J. Bus. Venturing*, vol. 23, no. 4, pp. 405–422, 2008.
- [25] T. Stuart and O. Sorenson, "The geography of opportunity: Spatial heterogeneity in founding rates and the performance of biotechnology firms," *Res. Policy*, vol. 32, no. 2, pp. 229–253, 2003.
- [26] A. Saxenian, "Inside-out: Regional networks and industrial adaptation in Silicon Valley and Route 128," *Citiescape*, vol. 2, pp. 41–60, 1996.
- [27] T. J. Allen, O. Raz, and P. A. Gloor, "Does geographic clustering still benefit high tech new ventures? The case of the Cambridge/Boston biotech cluster," MIT ESD-WP-2009-01 Working Paper, 2009.
- [28] D. B. Audretsch and M. P. Feldman, "Knowledge spillovers and the geography of innovation," in *Handbook of Regional and Urban Economics*, vol. 4. Amsterdam, The Netherlands: Elsevier, 2004, pp. 2713–2739.
- [29] S. B. Letaifa and Y. Rabeau, "Too close to collaborate? How geographic proximity could impede entrepreneurship and innovation," *J. Bus. Res.*, vol. 66, no. 10, pp. 2071–2078, 2013.
- [30] M. E. Porter and M. P. Porter, "Location, clusters, and the 'New' microeconomics of competition," *Bus. Econ.*, vol. 33, pp. 7–13, 1998.
- [31] J. Singh, "Collaborative networks as determinants of knowledge diffusion patterns," *Manage. Sci.*, vol. 51, no. 5, pp. 756–770, 2005.
- [32] A. Al-Laham and V. Souitaris, "Network embeddedness and new-venture internationalization: Analyzing international linkages in the German biotech industry," *J. Bus. Venturing*, vol. 23, no. 5, pp. 567–586, 2008.
- [33] D. B. Audretsch and M. Keilbach, "The theory of knowledge spillover entrepreneurship," *J. Manage. Stud.*, vol. 44, no. 7, pp. 1242–1254, 2007.
- [34] M. McAdam and S. Marlow, "Building futures or stealing secrets? Entrepreneurial cooperation and conflict within business incubators," *Int. Small Bus. J.*, vol. 25, no. 4, pp. 361–382, 2007.
- [35] E. G. Carayannis *et al.*, "Technological learning for entrepreneurial development (TL4ED) in the knowledge economy (KE): Case studies and lessons learned," *Technovation*, vol. 26, no. 4, pp. 419–443, 2006.
- [36] G. Albort-Morant and D. Ribeiro-Soriano, "A bibliometric analysis of international impact of business incubators," *J. Bus. Res.*, vol. 69, no. 5, pp. 1775–1779, 2016.
- [37] G. G. Udell, "Are business incubators really creating new jobs by creating new business and new products," *J. Product Innov. Manage.*, vol. 7, no. 2, pp. 108–122, 1990.
- [38] D. A. Lewis, *Does Technology Incubation Work? A Critical Review*, Washington, DC, USA: Econ. Develop. Admin., U.S. Dept. Commerce, 2001.
- [39] G. Avnimelech, D. Schwartz, and R. Bar-El, "Entrepreneurial high-tech cluster development: Israel's experience with venture capital and technological incubators," *Eur. Plan. Stud.*, vol. 15, no. 9, pp. 1181–1198, 2007.
- [40] D. M. De Carolis and P. Saparito, "Social capital, cognition, and entrepreneurial opportunities: A theoretical framework," *Entrepreneurship Theory Pract.*, vol. 30, no. 1, pp. 41–56, 2006.
- [41] D. De Clercq and P. Arenius, "Effects of human capital and social capital on entrepreneurial activity," in *Proc. Babson Kauffman Entrepreneurship Res. Conf.*, 2002, pp. 157–170.
- [42] S. Y. Cooper and J. S. Park, "The impact of 'incubator' organizations on opportunity recognition and technology innovation in new, entrepreneurial high-technology ventures," *Int. Small Bus. J.*, vol. 26, no. 1, pp. 27–56, 2008.
- [43] R. Ferguson and C. Olofsson, "Science parks and the development of NTBFs—Location, survival and growth," *J. Technol. Transfer*, vol. 29, no. 1, pp. 5–17, 2004.
- [44] L. Pittaway *et al.*, "Networking and innovation: a systematic review of the evidence," *Int. J. Manage. Rev.*, vol. 5, nos. 3/4, pp. 137–168, 2004.
- [45] J. C. Hayton, "Competing in the new economy: the effect of intellectual capital on corporate entrepreneurship in high-technology new ventures," *R&D Manage.*, vol. 35, no. 2, pp. 137–155, 2005.
- [46] M. Battisti and M. McAdam, "Challenges of social capital development in the university science incubator The case of the graduate entrepreneur," *Int. J. Entrepreneurship Innov.*, vol. 13, no. 4, pp. 261–276, 2012.
- [47] E. Shaw, "Small firm networking: An insight into contents and motivating factors," *Int. Small Bus. J.*, vol. 24, no. 1, pp. 5–29, 2006.
- [48] E. T. Tornikoski and S. L. Newbert, "Exploring the determinants of organizational emergence: A legitimacy perspective," *J. Bus. Venturing*, vol. 22, no. 2, pp. 311–335, 2007.
- [49] P. S. Adler and S.-W. Kwon, "Social capital: Prospects for a new concept," *Acad. Manage. Rev.*, vol. 27, no. 1, pp. 17–40, 2002.
- [50] T. Kawakami, S. S. Durmuşoğlu, and G. Barczak, "Factors influencing information technology usage for new product development: The case of Japanese companies," *J. Product Innov. Manage.*, vol. 28, no. 6, pp. 833–847, 2011.
- [51] M. Lorenzen, "Social capital and localised learning: proximity and place in technological and institutional dynamics," *Urban Stud.*, vol. 44, no. 4, pp. 799–817, 2007.

- [52] J. Liao and H. Welsch, "Social capital and entrepreneurial growth aspiration: a comparison of technology- and non-technology-based nascent entrepreneurs," *J. High Technol. Manage. Res.*, vol. 14, no. 1, pp. 149–170, 2003.
- [53] A. Verhoeff and H. Menzel, "Social capital to bridge the valley of death, simulating critical incidents in innovation," *Int. J. Entrepreneurship Small Bus.*, vol. 14, no. 1, pp. 149–169, 2011.
- [54] A. Robb and J. Farhat, *An Overview of the Kauffman Firm Survey: Results From 2011 Business Activities*. Kansas City, MO, USA: Ewing Marion Kauffman Found., 2013.
- [55] J. B. Farhat and A. Robb, *Applied Survey Data Analysis Using Stata: The Kauffman Firm Survey Data*, 2014. [Online]: Available: SSRN 2477217.
- [56] M. Granovetter, "The strength of weak ties: A network theory revisited," *Sociol. Theory*, vol. 1, no. 1, pp. 201–233, 1983.
- [57] M. Granovetter, "The strength of weak ties," *Amer. J. Sociol.*, vol. 78, no. 6, pp. 1360–1380, 1973.
- [58] D. Li, "Multilateral R&D alliances by new ventures," *J. Bus. Venturing*, vol. 28, no. 2, pp. 241–260, 2013.
- [59] A. R. Anderson and S. L. Jack, "The articulation of social capital in entrepreneurial networks: A glue or a lubricant?" *Entrepreneurship Reg. Develop.*, vol. 14, no. 3, pp. 193–210, 2002.
- [60] L. Pirolo and M. Presutti, "The impact of social capital on the start-ups' performance growth," *J. Small Bus. Manage.*, vol. 48, no. 2, pp. 197–227, 2010.
- [61] A. Greve and J. W. Salaff, "Social networks and entrepreneurship," *Entrepreneurship Theory Pract.*, vol. 28, no. 1, pp. 1–22, 2003.
- [62] F. Liñán and F. J. Santos, "Does social capital affect entrepreneurial intentions?" *Int. Adv. Econ. Res.*, vol. 13, no. 4, pp. 443–453, 2007.
- [63] B. Efron, "The efficiency of Cox's likelihood function for censored data," *J. Amer. Statist. Assoc.*, vol. 72, no. 359, pp. 557–565, 1977.
- [64] D. Pregibon, "Goodness of link tests for generalized linear models," *J. Royal Stat. Soc., Ser. C (Appl. Statist.)*, vol. 29, no. 1, pp. 15–14, 1980.
- [65] J. M. Wooldridge, *Econometric Analysis of Cross Section and Panel Data*. Cambridge, MA, USA: MIT Press, 2010.
- [66] R. A. Baron and G. D. Markman, "Beyond social capital: How social skills can enhance entrepreneurs' success," *Acad. Manage. Executive*, vol. 14, no. 1, pp. 106–116, 2000.
- [67] P. Koellinger, M. Minniti, and C. Schade, "'I think I can, I think I can': Overconfidence and entrepreneurial behavior," *J. Econ. Psychol.*, vol. 28, no. 4, pp. 502–527, 2007.
- [68] M. Simon, S. M. Houghton, and K. Aquino, "Cognitive biases, risk perception, and venture formation: How individuals decide to start companies," *J. Bus. Venturing*, vol. 15, no. 2, pp. 113–134, 2000.
- [69] G. Xiong and S. Bharadwaj, "Social capital of young technology firms and their IPO values: The complementary role of relevant absorptive capacity," *J. Marketing*, vol. 75, no. 6, pp. 87–104, 2011.
- [70] M. Kauffeld-Monz, "Knowledge spillovers within regional networks of innovation and the contribution made by public research," *IUP J. Knowl. Manage.*, vol. 7, nos. 3/4, pp. 41–63, 2009.
- [71] H. Zhao and S. E. Seibert, *The Big Five Personality Dimensions and Entrepreneurial Status: A Meta-Analytical Review*. Washington, DC, USA: Amer. Psychological Assoc., 2006.
- [72] H. Zhao, S. E. Seibert, and G. T. Lumpkin, "The relationship of personality to entrepreneurial intentions and performance: A meta-analytic review," *J. Manage.*, vol. 36, no. 2, pp. 381–404, 2010.
- [73] H. Brandstätter, "Personality aspects of entrepreneurship: A look at five meta-analyses," *Personality Individual Differences*, vol. 51, no. 3, pp. 222–230, 2011.



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